

# Comparative Study of Post-Hoc Methods vs Intrinsic Interpretability in Vision Models

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June 2026

## The "Black Box" Problem

- Deep learning models (CNNs, ViTs) have opaque decision-making processes.
- **Post-Hoc XAI**: Explainers applied *after* training (e.g., **SHAP** [?], **LIME** [?]).
- **Intrinsic XAI**: Models interpretable by design (e.g., **ProtoPNet** [?]).

- **SHAP**

$$\phi_i = \sum_S \frac{|S|!(n - |S| - 1)!}{n!} \Delta v$$

- **LIME**

$$\xi(x) = \arg \min_{g \in G} \mathcal{L} + \Omega(g)$$

- **GradCAM**

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left( \sum_k \alpha_k^c A^k \right)$$

- **ProtoPNet**

$$\text{sim} = \log \left( \frac{d^2 + 1}{d^2 + \epsilon} \right)$$

- **Infidelity**

$$\text{INFD} = \mathbb{E}_I \left[ \left( f(x) - f(x - I) - I^T \Phi(x) \right)^2 \right]$$

- **Complexity (Entropy)**

$$C(\Phi) = - \sum_i p_i \ln(p_i)$$

- **Localization (IoU)**

$$\text{Loc} = \frac{|M \cap B|}{|M \cup B|}$$

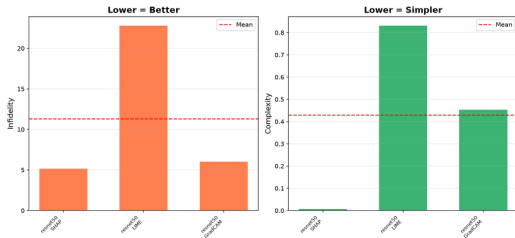
- **Rank Agreement (Spearman)**

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

# Experimental Setup & Metrics

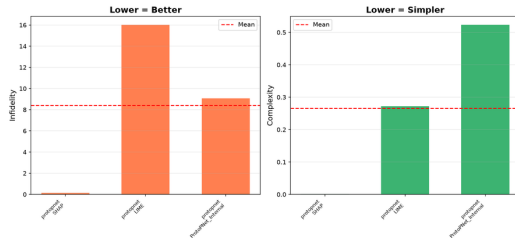
## Post-Hoc Metrics

XAI Metrics Comparison



## Intrinsic Metrics

XAI Metrics Comparison

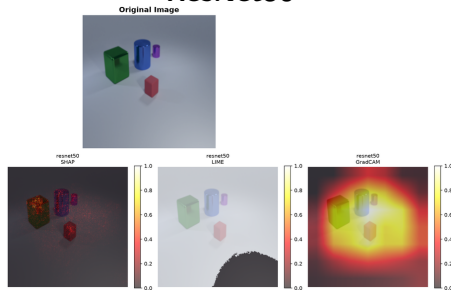


**Datasets:** *CLE4EVR* (3D objects) and *MNMath* (Handwritten equations).

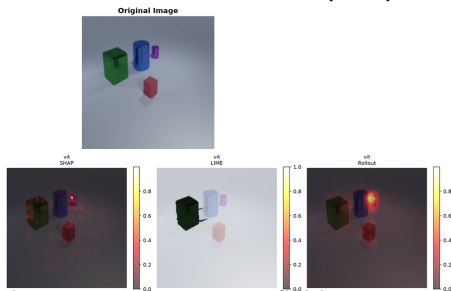
**Key Metrics:** *Infidelity*, *Complexity*, *Localization*.

# Post-Hoc Methods on CLE4EVR Dataset

## ResNet50



## Vision Transformer (ViT)

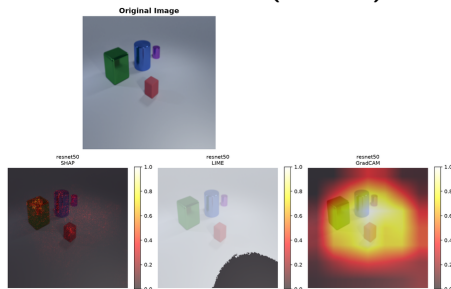


- **GradCAM** performs well: large 3D objects match its coarse receptive field.
- **LIME** underperforms: default SLIC superpixels severely fragment the smooth 3D shapes.

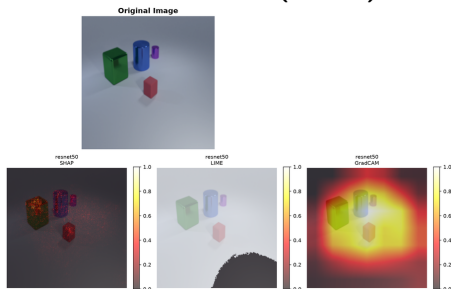
# LIME Ablation Study on CLE4EVR

- **Hypothesis:** Does matching LIME's superpixels ( $K$ ) to the number of objects improve explanations?
- **Result:** Object-aligned boundaries ( $K = 4$ ) unexpectedly double the *Infidelity* compared to default ( $K = 50$ ). Linear surrogates fail to capture the non-linear reasoning of CNNs.

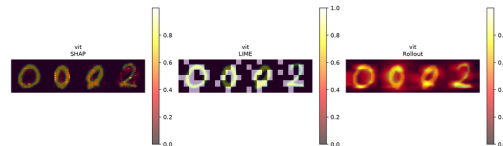
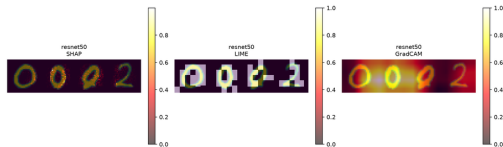
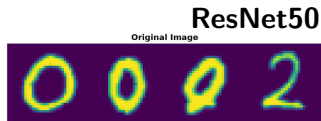
Default LIME ( $K = 50$ )



Ablated LIME ( $K \approx 4$ )

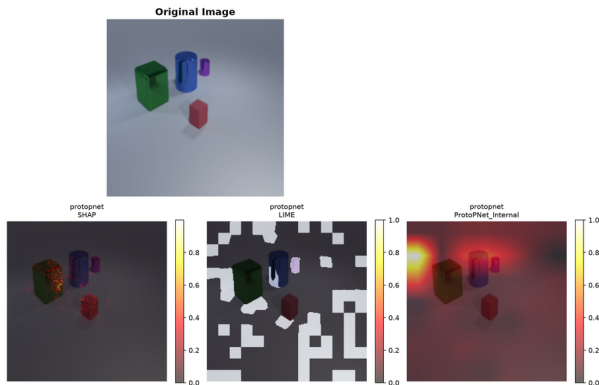


# Post-Hoc Methods on MNMath Dataset



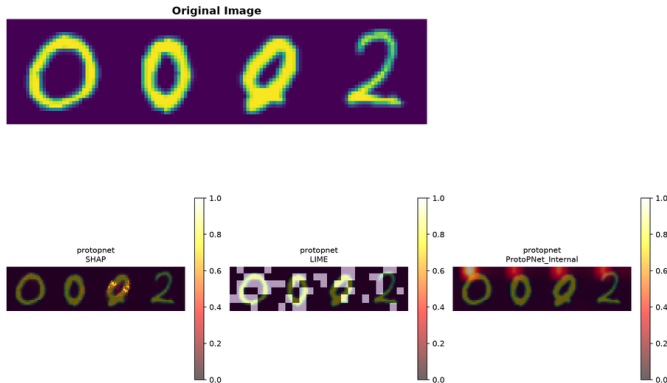
- **SHAP** heavily outperforms GradCAM on fine-grained stroke detection.
- GradCAM's low spatial resolution blurs over the thin handwritten digits.

## Visual Comparison on CLE4EVR



**SHAP** isolates edges (sparse), while **ProtoNet Internal** highlights broad, context-aware regions.

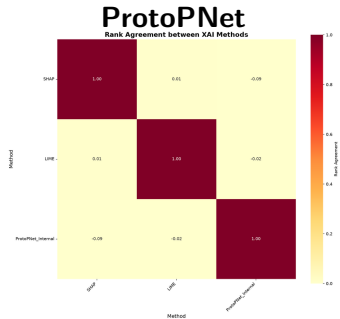
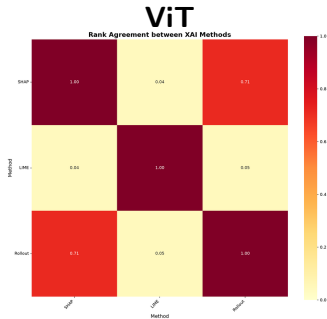
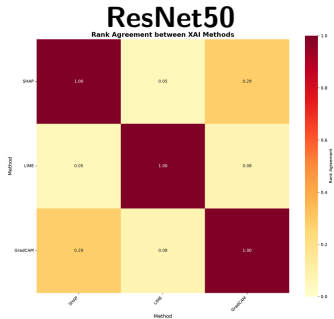
## Visual Comparison on MNMath



**ProtoPNet** reasoning encompasses the whole digit and its context, completely ignoring the strict "stroke" pixels favored by SHAP.

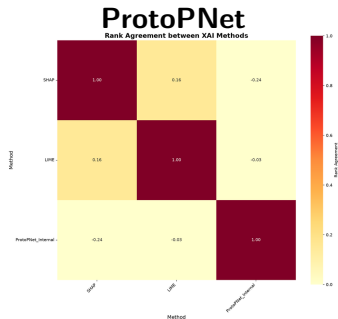
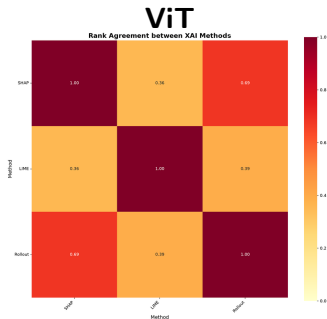
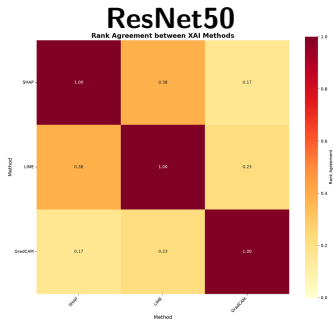
# Rank Agreement: CLE4EVR Dataset

- Correlation between Spatial Locality (Intersection) and Infidelity.
- Across all architectures, post-hoc rank correlation is **near zero or negative**.



# Rank Agreement: MNMATH Dataset

- The disconnect is even worse on fine-grained data (-0.24 for ProtoPNet).
- Post-hoc methods misrepresent the actual broad spatial reasoning of the network!



## Key Takeaways

- Relying strictly on "Localization Accuracy" is dangerous: it assumes networks reason using human-defined object boundaries.
- Intrinsic interpretability reveals that models learn broad spatial contexts, unlike what post-hoc methods suggest [?].

# Thank You!

Questions?

# References I